PROJECT REPORT

Lead Lifecycle & Conversion Intelligence System

Closed-deal performance analysis using Python, SQL and Power BI

CLOSED DEALS	SALES REPS	SEGMENTS	LEAD TYPES
842	**22**	**34**	**9**
records analyzed	distinct representatives	business segments	including unknown

Prepared as a portfolio project

This report documents the analysis of the Olist closed-deals dataset and the Power BI dashboard built to explore the composition and timing of closed deals. It is written for portfolio review and interview discussion.

Scope note
The available file contains closed deals only. It does not include the full population of unconverted leads or complete stage history, so this analysis cannot calculate true conversion rates, funnel drop-off, or conversion efficiency.

1. Executive Summary

The project analyzes 842 closed-deal records to understand when deals were closed, which lead types and business segments appear most often among closed deals, and how deal counts are distributed across sales representatives and business types. The analysis was completed in Python/Pandas, validated with SQL queries, and presented in a two-page Power BI report.

Key findings

Finding	Evidence	Interpretation
Peak month	April 2018: 207 closed deals	Highest monthly count in the observed period; this does not explain why the peak occurred.
Largest lead type	online_medium: 332 (39.43%)	Largest share of recorded closed deals, not a conversion rate.
Largest business type	reseller: 587 (69.71%)	Resellers dominate the business-type mix in this closed-deal file.
Largest segment	home_decor: 105 (12.47%)	Most common segment among closed deals.
Monthly pattern	Counts fell after April 2018	Trend warrants investigation; causes cannot be inferred from this dataset alone.

Project objectives

- Assess dataset quality and prepare analysis-ready fields.
- Summarize closed deals by month, lead type, business segment, business type and sales representative.
- Cross-check key aggregations in SQL.
- Build a Power BI dashboard for clear, interactive exploration.
- Document limitations so findings are not overstated.

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2. Dataset & Methodology

Source and scope

The analysis uses the Olist closed-deals dataset supplied for the project. The working table contains 842 records and 14 original columns; a derived month field was added for time-series analysis, producing a cleaned table with 15 columns.

Field group	Examples	Use in analysis
Identifiers	mql_id, seller_id, sdr_id, sr_id	Count unique deals and summarize representative activity.
Timing	won_date, won_month	Monthly closed-deal trend.
Commercial categories	business_segment, lead_type, business_type	Compare mix of recorded closed deals.
Business attributes	has_company, has_gtin, average_stock	Available for future segmentation; high missingness limits use.
Declared values	declared_monthly_revenue, declared_product_catalog_size	Potential business context; missingness and zero values require care.

Tools and workflow

1. Python/Pandas: load, inspect, clean categorical fields, create a month field, and perform descriptive summaries.
2. SQL (SQLite): register the cleaned dataframe as `closed_deals` and independently validate counts and category breakdowns.
3. Power BI Desktop: create KPI cards, a monthly trend line, and categorical charts for exploration.

Metric definitions

Closed deals = distinct `mql_id` records in this dataset. **Deal share** = a category's closed-deal count divided by 842, multiplied by 100. These are descriptive shares within the closed-deal sample; they are not lead conversion rates.

3. Data Quality & Preparation

The data was checked for missing values, duplicate rows, duplicate deal identifiers, and missing close dates. Categorical fields were trimmed and blank/null values were labeled “Unknown” where appropriate. The original row count was retained.

Quality check	Result	Action / implication
Rows	842 before and after cleaning	No records removed during the described cleaning step.
Duplicate rows	0	No exact duplicate rows detected.
Duplicate mql_id	0	Each recorded deal identifier is unique.
Missing won_date	0	All records can be assigned to a month.
Missing business_segment after cleaning	0	One missing segment was categorized as Unknown during cleaning, if applicable.

Important missingness

Column	Missing count	Missing share
has_company	779	92.52%
has_gtin	778	92.40%
average_stock	776	92.16%
declared_product_catalog_size	773	91.81%
lead_behaviour_profile	177	21.02%
business_type	10	1.19%
lead_type	6	0.71%

The high missingness in several business-attribute fields means analyses using those columns may be incomplete or biased. The 773 missing catalog-size values were not imputed. Declared monthly revenue has 797 zero values (94.66%); zeros were preserved rather than assumed to be missing.

4. Findings: Timing & Category Mix

4.1 Monthly closed-deal trend

The dataset covers December 2017 through November 2018. Recorded monthly counts rise to a peak in April 2018 and then decline through November. December 2017's baseline is only three deals, so the very large January month-over-month percentage is not a stable signal by itself.

Month	Closed deals	MoM change
Dec 2017	3	—
Jan 2018	73	+2333.33%
Feb 2018	113	+54.79%
Mar 2018	147	+30.09%
Apr 2018	207	+40.82%
May 2018	122	-41.06%
Jun 2018	57	-53.28%
Jul 2018	37	-35.09%
Aug 2018	33	-10.81%
Sep 2018	23	-30.30%
Oct 2018	21	-8.70%
Nov 2018	6	-71.43%

Interpretation

April 2018 recorded 207 closed deals, the maximum monthly count. The count then fell to 122 in May (-41.06%), 57 in June (-53.28%), and 6 in November (-71.43% from October). These changes describe the observed series; the data does not establish whether seasonality, sales-team changes, data coverage, or other factors caused the pattern.

4.2 Lead-type mix

Lead type	Closed deals	Share
online_medium	332	39.43%
online_big	126	14.96%
industry	123	14.61%
offline	104	12.35%
online_small	77	9.14%
online_beginner	57	6.77%
online_top	14	1.66%
Unknown	6	0.71%
other	3	0.36%

The three largest lead types—online_medium, online_big and industry—account for 581 deals, or 69.00% of the recorded closed-deal sample. This concentration can guide descriptive segmentation, but it does not show which lead type converts best because counts of unconverted leads are unavailable.

5. Findings: Business & Sales Mix

5.1 Business segments

Business segment	Closed deals	Share
home_decor	105	12.47%
health_beauty	93	11.05%
car_accessories	77	9.14%
household_utilities	71	8.43%
construction_tools_house_garden	69	8.19%
audio_video_electronics	64	7.60%
computers	34	4.04%
pet	30	3.56%
food_supplement	28	3.33%

The five leading segments account for 415 deals (49.28%). Home decor is the largest single segment in the recorded sample, followed by health and beauty and car accessories.

5.2 Business-type mix

Business type	Closed deals	Share	Sales reps represented
reseller	587	69.71%	21
manufacturer	242	28.74%	19
Unknown	10	1.19%	7
other	3	0.36%	3

Resellers represent about seven in ten closed-deal records. This describes the composition of recorded deals, not relative business value or propensity to close.

5.3 Sales representative distribution

Sales representative ID (shortened)	Closed deals	Deal share	Segments handled
4ef15afb…7afefe	133	15.80%	27
d3d1e91a…1955e3	82	9.74%	26
6565aa9c…7af3a9ba	74	8.79%	23
85fc447d…199fbc8	64	7.60%	—
495d4e95…2aa328	63	7.48%	—

There are 22 distinct sales representatives. The highest-count representative is associated with 133 closed deals (15.80%). Differences in counts should not be interpreted as performance rankings without workload, tenure, lead allocation, territory, and opportunity-quality context.

6. Power BI Dashboard

The Power BI report contains two analytical pages. Page 1 is an executive overview; Page 2 focuses on business and sales mix. Together, they provide a concise summary while allowing the viewer to inspect category distributions.

Page 1 — Executive Overview

• KPI cards: total closed deals (842), sales representatives (22), business segments (34), and lead types (9).
• Monthly closed-deal line chart for December 2017–November 2018.
• Horizontal bar chart of closed deals by lead type.
• Horizontal bar chart of top business segments by closed deals.
• Month range slicer for focused exploration.

Page 2 — Lead & Business Intelligence

• KPI cards for reseller deals (587), manufacturer deals (242), top lead type (online_medium), and top business segment (home_decor).
• Donut chart showing the business-type mix.
• Stacked horizontal bar chart comparing business segments by business type.
• Bar chart of the top 10 sales representatives by closed deals.
• Clustered column chart of lead type by business type.

Design assessment

The dashboard separates high-level trends from category and representative detail, uses clear titles, and presents counts directly. For a stronger portfolio presentation, keep chart titles explicit about the metric (“Closed deals” rather than “Conversion”), use consistent number formatting, ensure labels remain legible at normal viewing size, and keep the slicer aligned with the chart it filters.

Recommended dashboard refinements

1. Add a visible scope note: “Closed-deal records only; conversion rate not available.”
2. Consider a compact table with segment, closed deals and share for precise comparison.
3. Use a tooltip to show exact counts and percentages where labels are crowded.
4. Use consistent colors for business types across all visuals.
5. Add a date range caption and verify that every visual responds to the month slicer as intended.

7. Recommendations & Next Steps

Business-facing recommendations

• Review the sharp fall in monthly closed-deal counts after April 2018 with additional operational data before attributing a cause.
• Use the observed lead-type and segment mix to plan descriptive reporting and stakeholder questions, not to claim one source is more effective.
• Review sales-representative workload and opportunity allocation before comparing counts as performance.
• Improve data completeness for company, GTIN, stock and catalog-size fields if these attributes are important to business decisions.
• Confirm what zero means in declared monthly revenue with the data owner before using it as a numeric business KPI.

Data required to measure actual conversion

To extend this project into a true lead lifecycle/conversion analysis, obtain a dataset containing all leads—not only closed deals—with a unique lead identifier, lead creation date, source/type, stage timestamps, outcome/status, and close date where applicable. Then calculate stage-to-stage conversion rates, lead-to-deal conversion rate, time-to-close, and cohort performance using a clear denominator.

Suggested future KPIs (only after the required data is available)

KPI	Required definition/data
Lead-to-deal conversion rate	Closed leads divided by all eligible leads in a defined cohort.
Stage conversion rate	Leads entering the next stage divided by leads entering the prior stage.
Time to close	Difference between lead creation date and won date, with clear treatment of open leads.
Lead source effectiveness	Conversion and value by source, including both won and not-won leads.
Revenue contribution	Verified revenue or deal value linked to each opportunity; do not substitute declared prospect revenue.

8. Limitations & Responsible Interpretation

• Closed-deal-only selection: the dataset does not include all leads, so conversion rates, unsuccessful-lead comparisons and funnel drop-off cannot be calculated.
• Descriptive, not causal: month-to-month changes and category differences do not explain why outcomes occurred.
• Unequal exposure: representative counts may reflect differences in assignment, tenure, territory or lead quality.
• Missing attributes: several columns have over 90% missingness, limiting their reliability for segmentation.
• Declared revenue: the field appears to describe declared monthly revenue associated with the business, not revenue earned from the recorded deals; its meaning should be confirmed before use.

9. Conclusion

This project delivers a reproducible descriptive analysis of 842 closed-deal records, validates key summaries with SQL, and communicates the findings through a two-page Power BI dashboard. It identifies the peak recorded month, the leading lead type and business segment, the dominance of resellers in the recorded deal mix, and the distribution of closed deals across representatives.

The most important analytical discipline in this project is scope: the results describe closed deals in the available file. They do not establish lead conversion efficiency. With complete lead and stage-history data, the project could be extended into a genuine lifecycle funnel and conversion-performance analysis.

Glossary

Term	Meaning in this report
Closed deal	A record present in the supplied closed-deals dataset, counted by unique mql_id.
Deal share	Category closed-deal count divided by the 842 total records, expressed as a percentage.
Lead type	The categorical lead-type field provided in the dataset.
Business segment	The segment classification provided in the dataset, such as home_decor or health_beauty.
MoM change	Percentage change in the monthly closed-deal count relative to the immediately preceding month.
Unknown	A label used for missing or blank categorical values during cleaning; it is not an original business category.

End of report

Project: Lead Lifecycle & Conversion Intelligence System
Analysis tools: Python / Pandas · SQL / SQLite · Microsoft Power BI Desktop